

Econometrics: Methods and Applications

Test Exercise 3

Name: Vasileios Dimitropoulos

GitHub repo: [BillDimitro/econometrics-methods-and-applications](https://github.com/BillDimitro/econometrics-methods-and-applications)

Note: This PDF presents the final numerical results and interpretation for Test Exercise 3. All statistical tests and numerical computations were implemented in Python. The corresponding source code for this exercise is available in Module_4/scripts/test_ex_3.py, which is included in the GitHub repository referenced above. The code is extensively documented with comments explaining the methodology, implementation, and interpretation of the results. The repository also contains the training exercises and supporting work completed throughout this module.

(d) General-to-specific approach: We start by regressing the federal funds rate on all 7 available variables.

After each regression, we look at all p-values and eliminate the variable with the largest p-value (or the smallest absolute t-statistic), among those with $p > 0.05$.

We always keep the constant in the model to ensure the residuals have zero mean. This process is implemented in the function "gen-to-spec".

General Model

$$\text{INTRATE}_i = \beta_0 + \beta_1 \text{INFL}_i + \beta_2 \text{PROD}_i + \beta_3 \text{UNEMPL}_i + \beta_4 \text{COMPRI}_i + \beta_5 \text{PCE}_i + \beta_6 \text{PERSINC}_i + \beta_7 \text{HOUST}_i$$

Python

Regression 1:

Removing UNEMPL ($p\text{-value} = 0.290$)

Regression 2:

Removing PROD ($p\text{-value} = 0.323$)

Final Model:

$$\text{INTRATE}_i = \beta_0 + \beta_1 \text{INFL}_i + \beta_2 \text{COMPRI}_i + \beta_3 \text{PCE}_i + \beta_4 \text{PERSINC}_i + \beta_5 \text{HOUST}_i + \epsilon_i$$

Final Regression Statistics:

Variable	β	t-statistic	p-value
const	-0.24	-1.042	0.298
INFL	0.718	12.555	<0.001
COMPRI	-0.008	-2.841	0.005
PCE	0.341	5.756	<0.001
PERSINC	0.24	4.048	<0.001
HOUST	-0.021	-4.678	<0.001

Model fit: $R^2 = 0.637$

(b) Specific-to-general approach: We start by regressing the federal funds rate on only a constant, and then add 1 variable at a time. At each iteration, we perform as many regressions as the available variables, we find the variable with the smallest p-value (or the largest absolute t-statistic), and we add it to the model only if $p < 0.05$. We then remove the added variable from the available variables list, and we repeat the process until all remaining variables satisfy $p < 0.05$.

This process is implemented in the function "spec-to-gen".

Starting Model

$$\text{INTRATE}_i = \beta_0 + \epsilon_i$$

⇓ Python

Iteration 1

Selected variable is INFL

Iteration 2:

Selected variable is PERSINK

Iteration 3:

Selected variable is PCE

Iteration 4:

Selected variable is HOUST

Iteration 5:

Selected variable is COMMPRI

Final Model

$$\text{INTRATE}_i = \beta_0 + \beta_1 \text{INFL}_i + \beta_2 \text{PERSINC}_i + \beta_3 \text{PCE}_i + \beta_4 \text{HOUT}_i + \beta_5 \text{COMMPRI}_i$$

Coefficients, t -statistics, p -values and R -squared match exactly the numbers of the general-to-specific approach.

While the selection paths differ because the significance of a variable depends on the other variables already included in the model, the specific-to-general approach leads to the same final model as the general-to-specific approach of (a).

c)

Akaike information criterion: $\text{AIC} = \log(s^2) + \frac{2K}{n}$

Bayes information criterion: $\text{BIC} = \log(s^2) + \frac{K \log(n)}{n}$

Both information criteria were computed within the "mult-reg" function and returned as part of its output.

Taylor rule:

$$\text{INTRATE}_i = \beta_0 + \beta_1 \text{INFL}_i + \beta_2 \text{PROD}_i$$

Model Comparison

	Gen-to-spec	Taylor rule
R^2	0.637	0.575
AIC	1.584	1.730
BIC	1.625	1.750

The general-to-specific model is preferred over the simple Taylor rule specification. It provides a better fit to the data ($R^2 = 0.637$ versus 0.575) and achieves lower AIC and BIC values. Therefore, the improvement in explanatory power more than compensates for the increase in model complexity.

d) The RESET test is implemented in the function "RESET-test".

Using $p=1$, the RESET test gives a p-value = $0.112 > 0.05$, so we do not reject the null of correct specification at the 5% level.

However, we went a step further and also tested with $p=2$, which yields a p-value = $0.105 > 0.05$, so it also does not reject. For $p=3$, the test gives p-value = $0.006 < 0.05$, suggesting possible higher order

nonlinearities. Since the exercise does not specify the RESET order, we base the main conclusion on $p=1$.

→ The Chow break test was implemented in the function "Chow-break-test".

It yields an F-statistic = 28.735 with a p-value that is effectively zero. Therefore, we reject the null hypothesis of parameter stability across the two subsamples. There is strong evidence of a structural break in the Taylor rule at 1980.

→ The Chow forecast test was implemented in the function "Chow-forecast-test". It yields an F-statistic = 5.511 with a p-value of essentially zero. Therefore, we reject the null of $r=0$, which means that the Taylor rule estimated on the pre-1980 sample does not adequately describe the post-1980 observations.

→ The Jarque-Bera test was implemented in the function "Jarque-Bera". It gives a JB-statistic = 12.526 and p-value = 0.002 < 0.05. Therefore, we reject the null hypothesis that the error terms are normally distributed. Combined with the Chow tests, this suggests that the Taylor rule model suffers from instability and non-normal residuals.